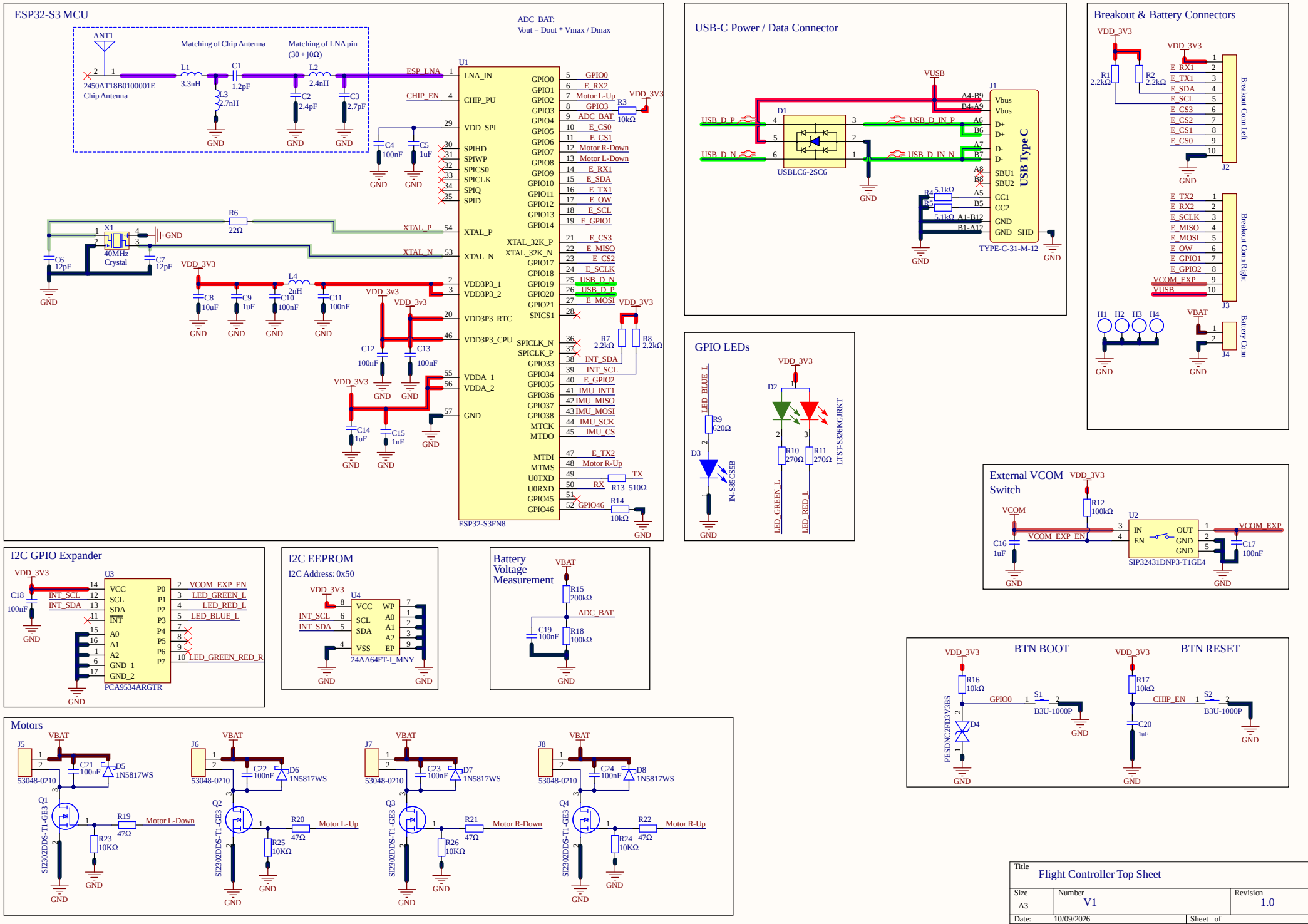
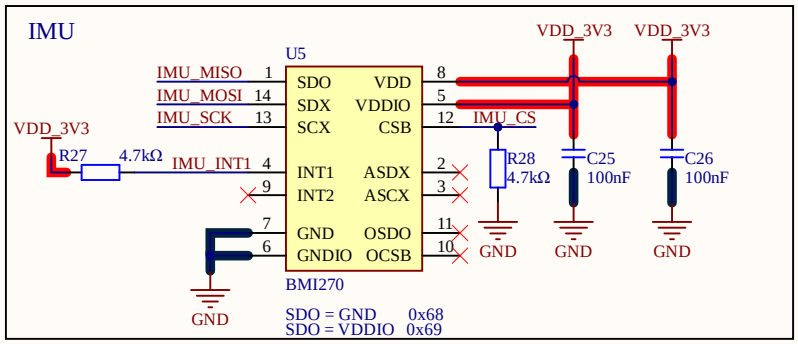
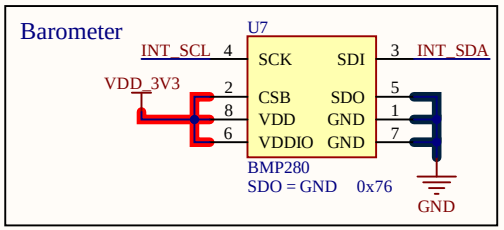
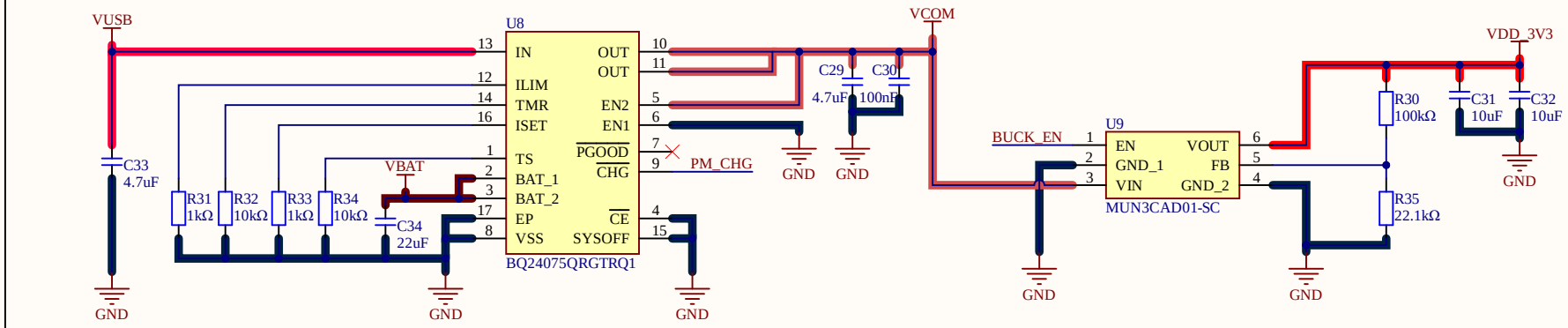

ESP32-S3 MCU
ADC_BAT:
Vout = Dout * Vmax / Dmax
Matching of Chip Antenna
Matching of LNA pin
(30 + j0Ω)
2450AT18B0100001E
Chip Antenna
ESP32-S3FN8
USB-C Power / Data Connector
USBLC6-2SC6
TYPE-C-31-M-12
USB Type C
Breakout & Battery Connectors
Breakout Conn Left
Breakout Conn Right
Battery Conn
GPIO LEDs
LED_BLUE_L
LED_GREEN_L
LED_RED_L
IN-S85CS5B
LTST-S326KGJRKT
External VCOM Switch
SIP32431DNP3-T1GE4
I2C GPIO Expander
PCA9534ARGTR
I2C EEPROM
I2C Address: 0x50
24AA64FT-I_MNY
Battery Voltage Measurement
BTN BOOT
BTN RESET
B3U-1000P
Motors
SI2302DDS-T1-GE3
1N5817WS
53048-0210
Flight Controller Top Sheet
Size A3
Number V1
Revision 1.0
Date: 10/09/2026
File: C:\Users\..\main.SchDoc
Sheet of
Drawn By: Cosmo Esposito





Title: Sensors		
Size: A4	Number: V1	Revision: 1.0
Date: 10/09/2026		Sheet of
File: C:\Users\..\sensors.SchDoc		Drawn By: Cosmo Esposito

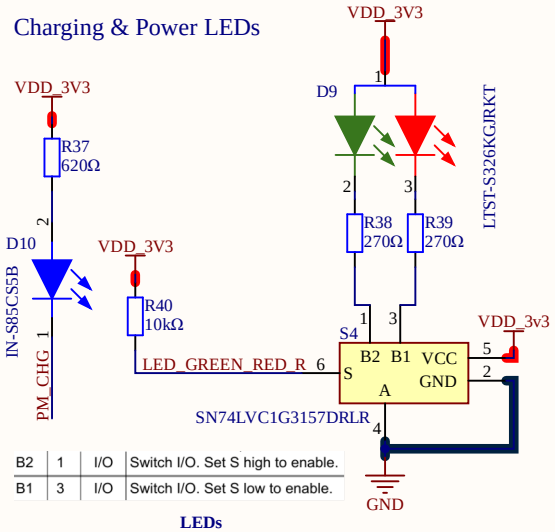
PMIC & 3.3V Buck Converter



R_{ISET} = 890 AΩ / I_{CHG} Fast charge current: 0.95A

R_{ILIM} = 1550AΩ / I_{I_MAX} Input current limit: 1.55A

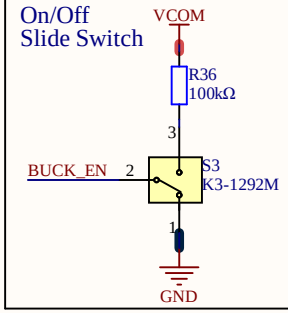
Charging & Power LEDs



B2	1	I/O	Switch I/O. Set S high to enable.
B1	3	I/O	Switch I/O. Set S low to enable.

LEDs
Default: Green
GPIO control to switch between Green & Red

On/Off Slide Switch



Title	Power Management		
Size: A4	Number: V1	Revision: 1.0	
Date: 10/09/2026		Sheet of	
File: C:\Users\..\pm.SchDoc		Drawn By: Cosmo Esposito	